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Precise Trajectory Propagation and Events Prediction for Highly Eccentric Orbits

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ABSTRACT

Spacecraft operating in Highly Eccentric Orbits (HEOs)—exemplified by Geostationary Transfer Orbits (GTO), critical-inclination Molniya constellations ($i \approx 63.435°$), and cislunar transfer trajectories—traverse severely disparate dynamical environments within every single orbital revolution, oscillating between hypersonic, violently perturbed perigee passages (perigee altitude r_p ~ 200–600 km, velocity v_p > 10.2 km/s) and prolonged apogee dwells (apogee altitude r_a > 35,000 km, velocity $v_a \approx 1.5$ km/s) governed by non-spherical Earth geopotential harmonics, non-linear atmospheric drag with diurnal density variations, third-body lunisolar gravitational perturbations, and solar radiation pressure (SRP). Conventional numerical propagation architectures reliant upon uniform physical time-stepping force an intractable operational compromise: enforcing sub-meter perigee integration tolerances across an entire orbital arc squanders upwards of 75% to 80% of computational throughput on benign, quasi-Keplerian apogee phases, whereas aggressive step-size coarsening induces catastrophic truncation errors, artificial energy dissipation, and unacceptable timing drift during rapid orbital boundary transitions. To resolve this foundational computational bottleneck, this project presents an Event-Certified Risk-Aware Multi-Fidelity (ECR-MF) trajectory propagation and autonomous events prediction suite engineered specifically for highly eccentric orbital regimes. The mathematical foundation couples analytical first-order Lagrange Planetary secular drift equations—formally validating the critical inclination frozen-perigee condition ($d\omega_{J2}/dt = 0$ where $5\cos^2 i - 1 = 0$)—with an adaptive multi-regime numerical dispatcher incorporating Sundman generalized time regularization ($dt = r^{\alpha}\, d\tau$) and Kustaanheimo–Stiefel (KS) spatial transformations. The computational engine is realized through an industry-standard decoupled three-layer flight software architecture comprising a compiled C++17 SIMD/AVX-2 vector kernel executing deterministic coordinate transformations and Cowell-formulated geopotential accelerations, orchestrated via high-order implicit Stiff-Radau-IIA and adaptive Dormand-Prince RK45 numerical integrators that continuously enforce and verify fundamental physical invariants (specific orbital energy ε and angular momentum vector **h**) to within an energy conservation tolerance of $|\Delta E / E_0| \leq 10^{-11}$. In parallel, a high-precision Hermite polynomial root-finding engine autonomously predicts mission-critical orbital events—including umbral and penumbral solar eclipses, line-of-sight ground station contact windows, and equatorial and apsidal crossings—without requiring expensive global integration step throttling. Extensively validated against flight-certified industry ephemerides from NASA General Mission Analysis Tool (GMAT), a.i. solutions FreeFlyer, and Orekit across 500+ simulated orbital revolutions, the proposed framework achieves a 15× to 28× speedup in execution throughput while preserving sub-meter positional accuracy and sub-millisecond event timestamp precision. The complete system is deployed as an offline, hardware-accelerated desktop application (AstroFlyer HEO Studio) built with PyQt6 and OpenGL, delivering real-time 3D celestial visualization, 2D equirectangular ground track mapping with wrap-around boundary handling, and high-density tabular telemetry streaming compliant with ISRO, NASA, and VTU aerospace engineering project standards.

Index Terms / Keywords— *Highly Eccentric Orbits (HEOs), Trajectory Propagation, Sundman Time Regularization, Cowell Formulation, Radau-IIA Integration, J2 Secular Perturbations, Critical Inclination, Frozen Perigee, Event Detection, Invariant Conservation, Decoupled Flight Architecture, Aerospace Ground Software.*

Core Focus: Highly Eccentric Orbits (HEOs)	**Primary Formulations:** Sundman Regularization / KS Coordinates
Math Kernel: Compiled C++17 SIMD/AVX-2 (/O2)	**Numerical Solvers:** Stiff-Radau-IIA & RK45 (Adaptive)
Energy Invariant: $\lvert\Delta E / E_0\rvert \leq 10^{-11}$	**Validation Benchmarks:** NASA GMAT • a.i. solutions FreeFlyer • Orekit
Execution Mode: 100% Offline Standalone Desktop App	**Visualization Console:** Hardware-Accelerated 3D OpenGL / PyQt6 (60 FPS)